<u>Project Report</u>

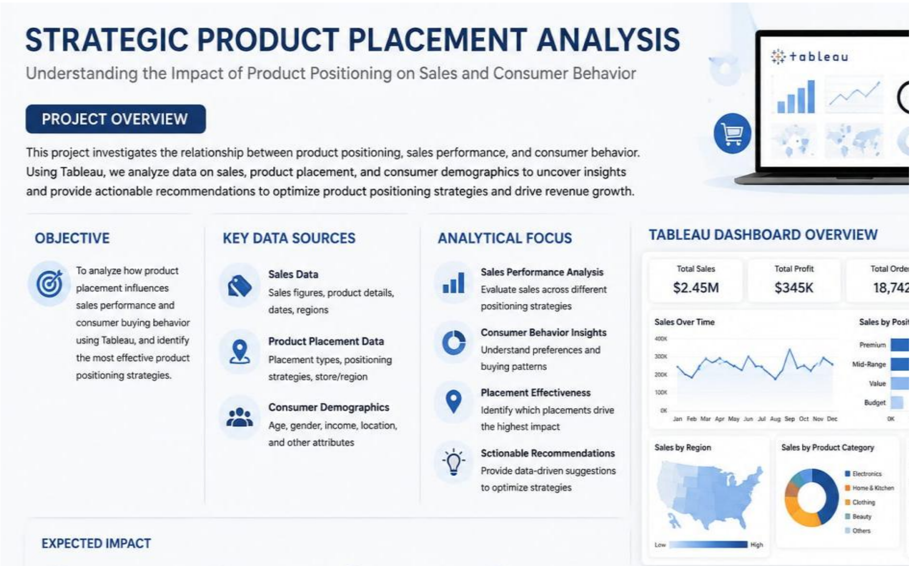
STRATEGIC PRODUCT PLACEMENT ANALYSIS

INTRODUCTION:

Project overview:

This project aims to investigate the relationship between product positioning, sales performance, and consumer behavior. Using Tableau, we will analyze data to uncover insights into how different positioning strategies impact sales and consumer preferences. By visualizing the data, we aim to provide actionable recommendations to optimize product positioning strategies and drive revenue growth.

A retail company wants to understand the impact of product positioning on its sales and consumer behavior. They have collected data on sales figures, product placement , and consumer demographics. They seek insights into which product positioning strategies are most effective in driving sales and how they can tailor their marketing efforts accordingly. Through data visualization with Tableau, the company hopes to gain actionable insights to improve its product positioning strategies and increase revenue.



Purpose:

The purpose of the Strategic Product Placement Analysis project is to analyze how product placement influences sales performance and customer buying behavior using Tableau. The project aims to identify the most effective product positioning strategies, uncover sales trends through data visualization, and provide insights that help businesses improve marketing decisions, optimize product placement, increase customer satisfaction, and maximize revenue growth.

IDEATION PHASE:

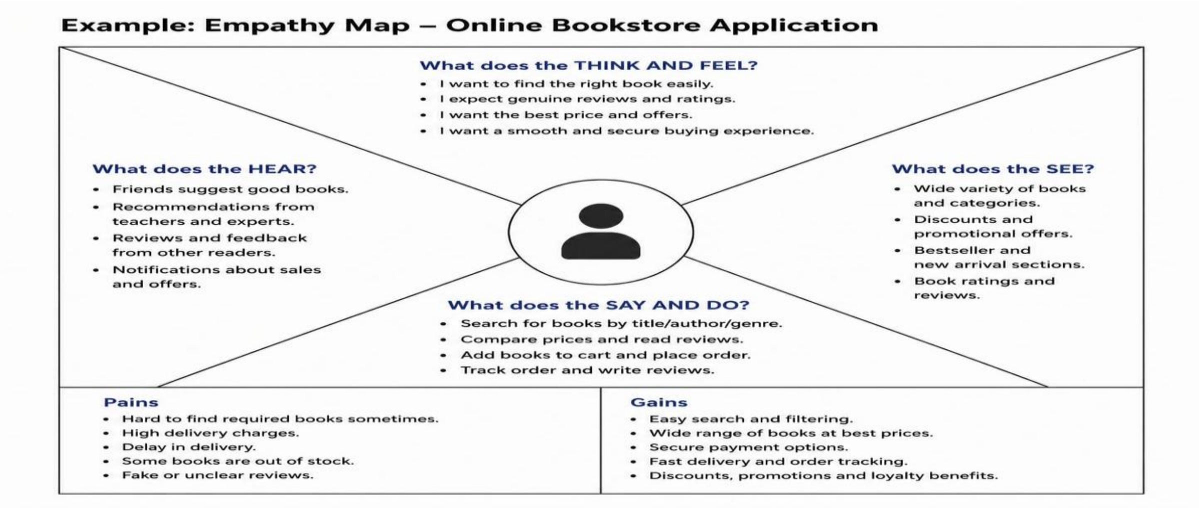
Problem Statement: Customers often face difficulty in finding products quickly because products are not placed strategically in stores. This leads to a poor shopping experience, wasted time, and reduced customer satisfaction. Example: Problem Statement (PS) I am (Customer) I'm trying to But Because PS-1 A retail store owner Improve product placement to increase sales I don't

I am	A retail customer who visits the store to purchase daily or weekly needed products.
I'm trying to	Find the products I need quickly and complete my shopping efficiently.
But	Products are not placed in the right locations, and relevant items are hard to find.
Because	The store does not use data-driven insights or customer behavior analysis to decide optimal product placement.
Which makes me feel	Frustrated and dissatisfied, and less likely to visit the store again.

Empathy Map Canvas:

An Empathy Map Canvas helps understand the needs, thoughts, feelings, and challenges of customers. For the Strategic Product Placement Analysis project, it is used to identify customer behaviour, shopping difficulties, and expectations. This helps retailers make better product placement decisions, improve the shopping experience, and increase sales..

Example:



BRAINSTROMING:

Brainstorming was conducted to identify the most effective solution for the Strategic Product Placement Analysis project. Various ideas such as analyzing product placement data, pre-processing the dataset, creating interactive Tableau dashboards, comparing product performance, and generating business recommendations were discussed. Based on impact and feasibility, developing an interactive Tableau dashboard was selected as the best solution because it provides clear visual insights and supports data-driven business decisions.

Step-1: Team Gathering, Collaboration and Select the Problem Statement

- Studied the domain of Strategic Product Placement Analysis.
- Identified the problem in analyzing the effectiveness of product placements.
- Selected the problem statement and defined the project objectives.

Step-2: Brainstorm, Idea Listing and Grouping

Ideas generated:

Data collection and pre-processing.

Product placement analysis.

Dashboard for visualization.

Report generation.

Step-3: Idea Prioritization

- After evaluating the ideas based on feasibility, accuracy, and project requirements, the following features were prioritized.
- Product placement analysis.
- Interactive dashboard.
- Report generation

REQUIREMENT ANALYSIS:

CUSTOMER JOURNEY MAP: This Map represents the Customer goes through while interacting through with a store (or online Platform) and how our System Supports each stage to improve product visibility, purchase decision and overall experience.

Stage	1. Awareness	2. Consideration	3. Decision	4. Purchase	5. Post-Purchase
Customer Action	Customer visits the store or online platform.	Customer browses products.	Customer selects a product.	Customer buys the product.	Customer gives feedback or reviews.
Touchpoint	Store, Website, Mobile App	Product Shelves, Search Results, Product Pages	Product Display, Recommendations	Billing Counter / Online Checkout	Feedback Form, App, Email, Review Section
Customer Goal	Find the required product.	Compare products and prices.	Choose the best product.	Complete the purchase.	Share experience and expectations.
System Response	Products are displayed strategically based on data insights.	System highlights popular, relevant and recommended products.	Strategic placement influences purchase decision.	Sales data is recorded for analysis.	System analyzes feedback and performance to improve future product placement.

SOLUTION REQUIREMENTS:

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	Data Upload	Upload dataset, Validate data
FR-2	Data Cleaning	Remove duplicates, Handle missing values
FR-3	Product Placement Analysis	Analyze product positioning and sales performance
FR-4	Dashboard	Displays charts, KPIs and filters
FR-5	Report Generation	Generate insights and recommendations

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

FR No.	Non-Functional Requirement	Description
NFR-1	**Usability**	The system should have a simple and user-friendly interface for easy navigation.
NFR-2	**Security**	User data should be protected with secure login and encrypted storage.
NFR-3	**Reliability**	The system should provide accurate product placement recommendations with minimal errors.
NFR-4	**Performance**	The application should load dashboards and analysis results within a few seconds.
NFR-5	**Availability**	The system should be available whenever users need it with minimal downtime.

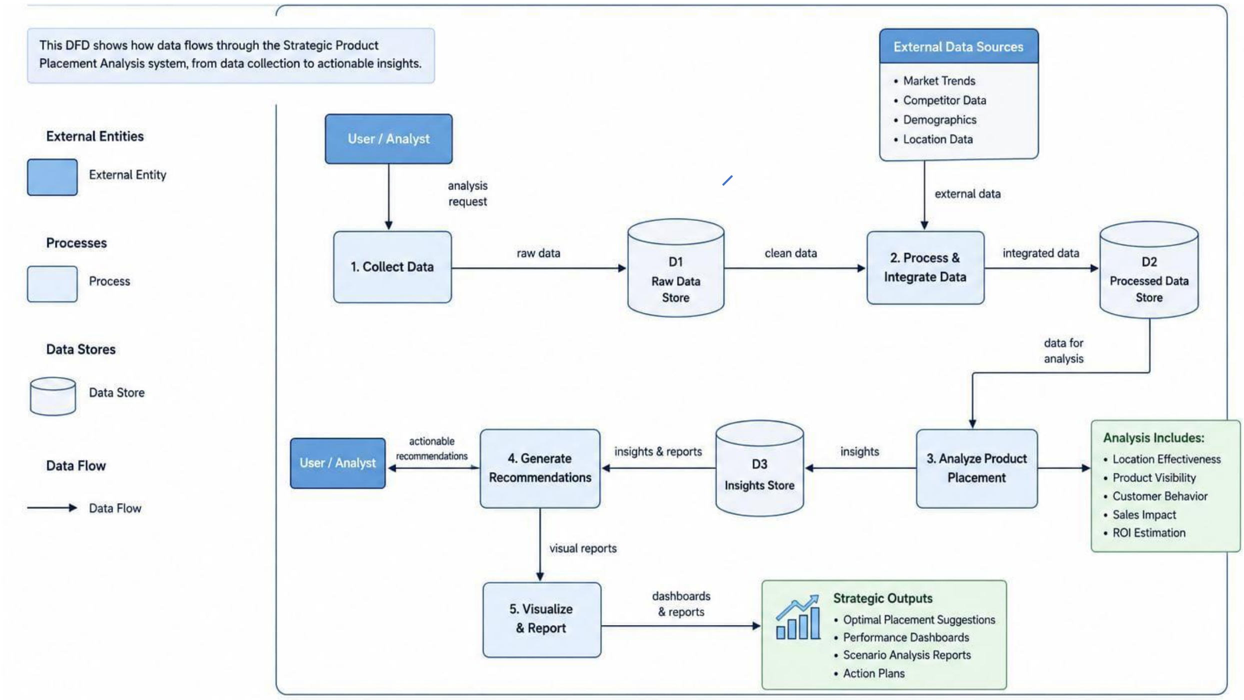
NFR-6	**Scalability**	The system should handle increasing numbers of users and larger datasets efficiently.

Data Flow Diagrams:

The Data Flow Diagram (DFD) illustrates how data moves through the Strategic Product Placement Analysis system. It shows the flow of information from the user to the system, where product, sales, and customer data are collected, validated, and processed. The system analyzes the data to identify the best product placement strategies and generates recommendations. Finally, the results are displayed as dashboards and reports to help businesses make informed decisions and improve product visibility and sales.

Example: (Simplified)

Example: Strategic Product Placement Analysis



TECHNOLOGY STACK:

Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2

The purpose of the Strategic Product Placement Analysis project is to analyze how product placement influences sales performance and customer buying behavior using Tableau. The project aims to identify the most effective product positioning strategies, uncover sales trends through data visualization, and provide insights that help businesses improve marketing decisions, optimize product placement, increase customer satisfaction, and maximize revenue growth.

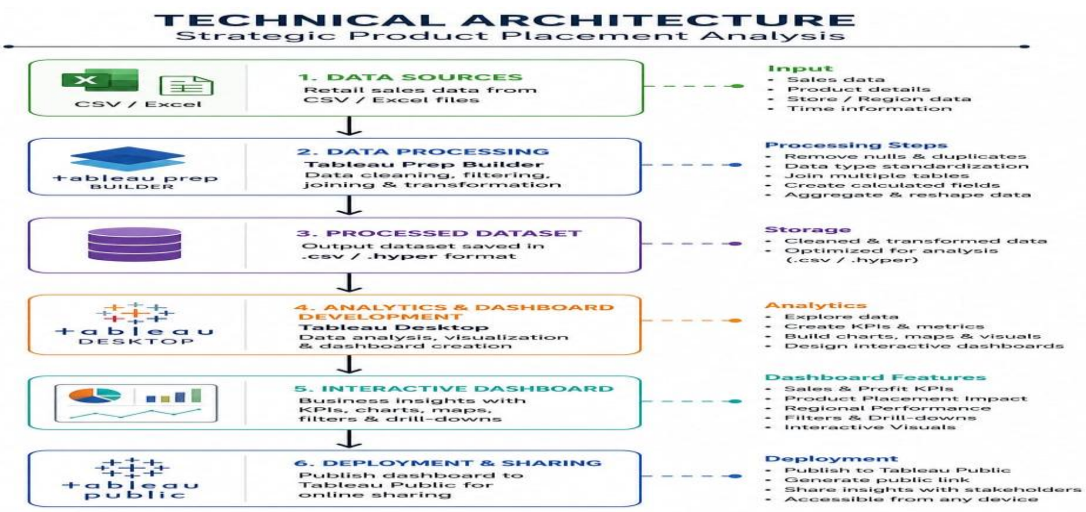


Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1.	User Interface	How user interacts with application e.g. Web UI, Mobile App, Chatbot etc.	HTML, CSS, JavaScript / Angular Js / React Js etc.
2.	Application Logic-1	Logic for a process in the application	Java / Python
3.	Application Logic-2	Logic for a process in the application	IBM Watson STT service
4.	Application Logic-3	Logic for a process in the application	IBM Watson Assistant
5.	Database	Data Type, Configurations etc.	MySQL, NoSQL, etc.
6.	Cloud Database	Database Service on Cloud	IBM DB2, IBM Cloudant etc.
7.	File Storage	File storage requirements	IBM Block Storage or Other Storage Service or Local Filesystem
8.	External API-1	Purpose of External API used in the application	IBM Weather API, etc.
9.	External API-2	Purpose of External API used in the application	Aadhar API, etc.
10.	Machine Learning Model	Purpose of Machine Learning Model	Object Recognition Model, etc.
11.	Infrastructure (Server / Cloud)	Application Deployment on Local System / Cloud Local Server Configuration: Cloud Server Configuration :	Local, Cloud Foundry, Kubernetes, etc.

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	List the open-source frameworks used	Technology of Opensource framework
2.	Security Implementations	List all the security / access controls implemented, use of firewalls etc.	e.g. SHA-256, Encryptions, IAM Controls, OWASP etc.

S.No	Characteristics	Description	Technology
3.	Scalable Architecture	Justify the scalability of architecture (3 – tier, Micro-services)	Technology used
4.	Availability	Justify the availability of application (e.g. use of load balancers, distributed servers etc.)	Technology used
5.	Performance	Design consideration for the performance of the application (number of requests per sec, use of Cache, use of CDN's) etc.	Technology used

PROJECT DESIGN:

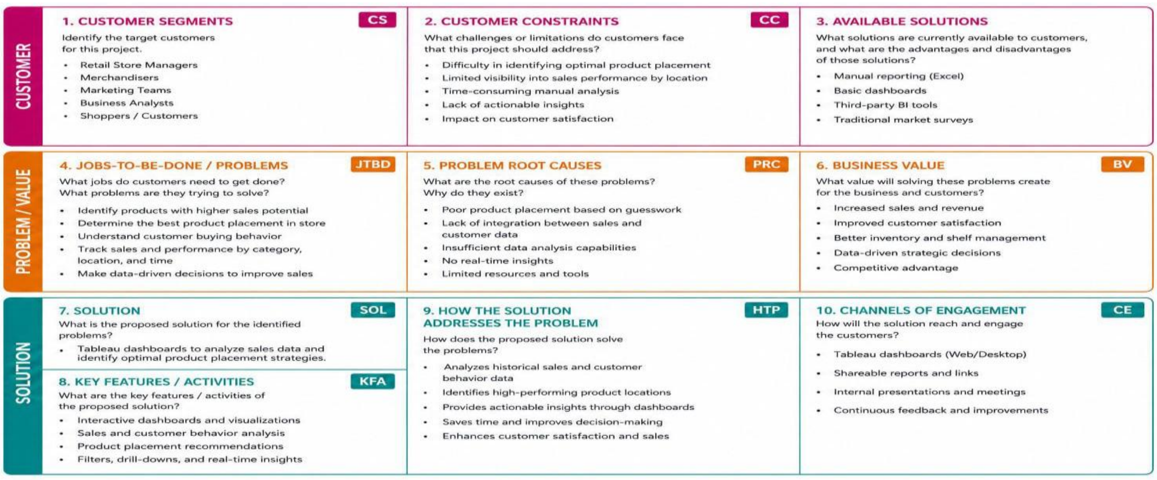
Problem – Solution Fit Template:

The Problem–Solution Fit for the Strategic Product Placement Analysis project identifies the challenges faced by retailers and customers due to poor product placement. The proposed solution uses sales data and customer purchasing patterns to recommend optimal product placement, improving customer satisfaction and increasing sales.

Purpose:

- Identify product placement problems in retail stores.
- Analyze customer buying behaviour using sales data.
- Recommend optimal product placement strategies.
- Improve customer shopping experience.
- Increase sales and support better business decisions.

Template:



Proposed Solution Template:

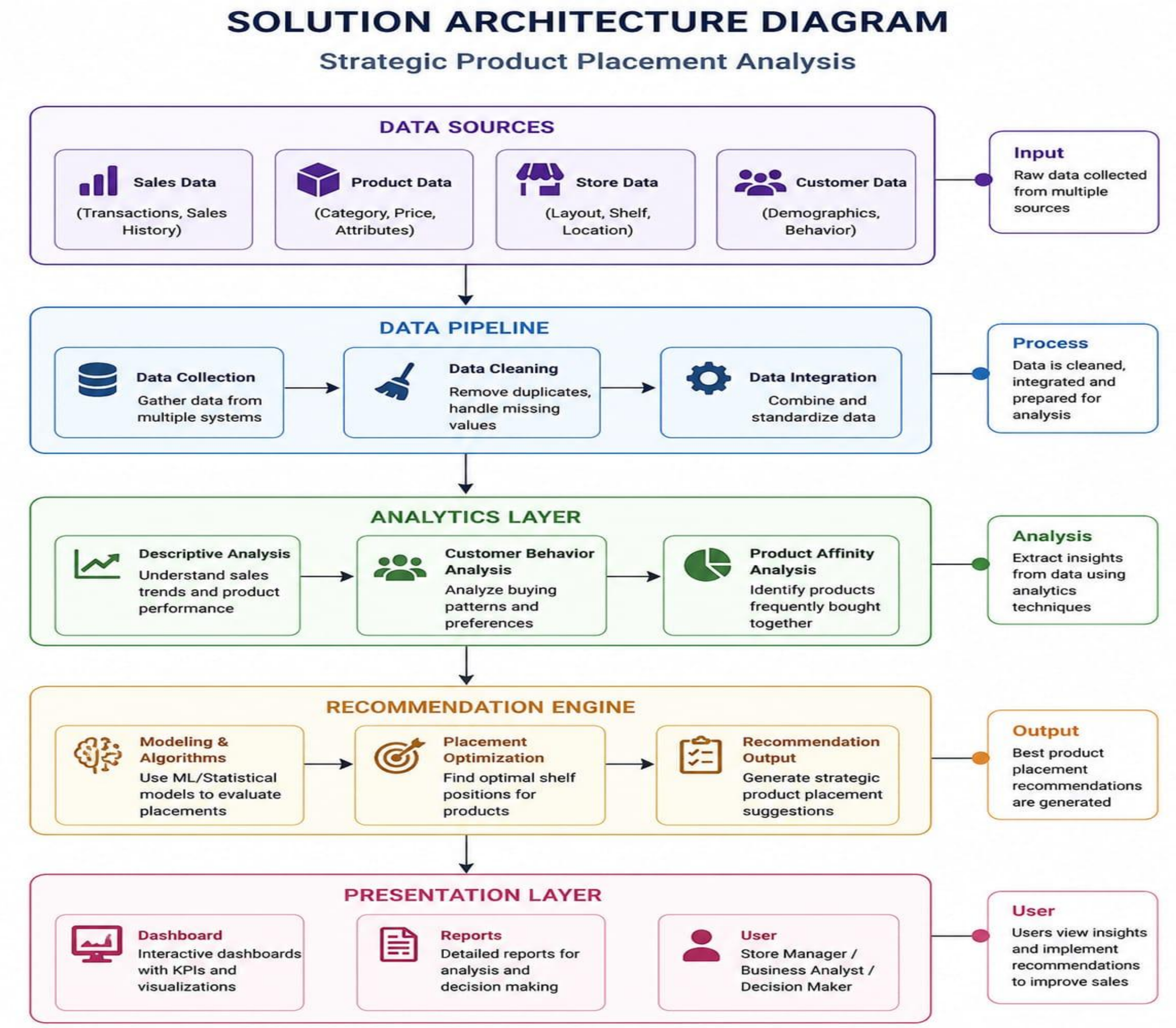
Project team shall fill the following information in the proposed solution template.

S.No.	Parameter	Description
1.	Problem Statement (Problem to be solved)	Retail store often place products without using customer purchase patterns, leading to poor product visibility and lower sales.
2.	Idea / Solution description	Develop a data-driven product placement analysis system that analyses sales data and recommends the best product arrangement to improve sales and customer convenience.
3.	Novelty / Uniqueness	Uses sales analytics and customer buying behaviour to recommend strategic product placement instead of relying on manual decisions.
4.	Social Impact / Customer Satisfaction	Helps customers find products easily, improves shopping experience, reduces search time, and increases customer satisfaction.
5.	Business Model (Revenue Model)	The system can be offered as a subscription (SaaS) or licensed to retail stores and supermarkets for improving store performance.
6.	Scalability of the Solution	The solution can be implemented in supermarkets, retail chains, grocery stores, and e-commerce businesses, and can scale to multiple store locations.

Solution Architecture:

The solution architecture of the Strategic Product Placement Analysis system illustrates how data flows through the application. It begins with collecting sales, customer, product, and store data. The data is then cleaned, processed, and analyzed to identify customer buying patterns and product performance. Based on these insights, the system generates recommendations for optimal product placement. Finally, the results are displayed through dashboards and reports to support business decision-making.

Example:



Collect sales, product, customer, and store data.

- Clean and integrate the collected data.
- Analyze customer behaviour and sales trends.
- Generate product placement recommendations.

- Display insights through dashboards and reports.

PROJECT PLANNING AND SHEDULING:

Product Backlog, Sprint Schedule, and Estimation (4 Marks)

Use the below template to create product backlog and sprint schedule

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
Sprint-1	Data Collection	USN-1	As a store manager, I want to collect sales data so that I can analyze customer purchases.	2	High	Self (Sana Pravalika)
Sprint-1	Data Cleaning	USN-2	As a user, I want to clean and organize data for accurate analysis	1	High	Self
Sprint-2	Customer Behaviour Analysis	USN-3	As a manager, I want to identify frequently purchased products	2	Low	Self
Sprint-1	Product Affinity analysis	USN-4	As a manager, I want product placement recommendations to increase sales.	2	Medium	Self
Sprint-1	Recommendation system	USN-5	As a user, I want to view the analysis results in a dashboard.	1	High	Self
	Dashboard &Report Generation		As a user, I want to view analysis results and reports in a dashboard.			Self

Project Tracker, Velocity & Burndown Chart: (4 Marks)

Sprint	Total Story Points	Duration	Sprint Start Date	Sprint End Date (Planned)	Story Points Completed (as on Planned End Date)	Sprint Release Date (Actual)
Sprint-1	20	6 Days	15 Feb 2025	20 Feb 2025	20	26 Feb 2025
Sprint-2	20	6 Days	21 Feb 2025	26 Feb 2025	20	26 Feb 2025
Sprint-3	20	6 Days	27 Feb 2025	4 Mar 2025	20	4 Mar 2025
Sprint-4	20	6 Days	5 Feb 2025	10 Mar 2025	20	10 Mar 2025

Velocity:
Imagine we have a 10-day sprint duration, and the velocity of the team is 20 (points per sprint). Let's calculate the team's average velocity (AV) per iteration unit (story points per day)

$$AV = \frac{sprint\ duration}{velocity} = \frac{20}{10} = 2$$

Burndown Chart:

The Burndown Chart represents the progress of the Strategic Product Placement Analysis project by showing the remaining tasks against the project timeline. It helps monitor project completion, identify delays, and ensure all activities.

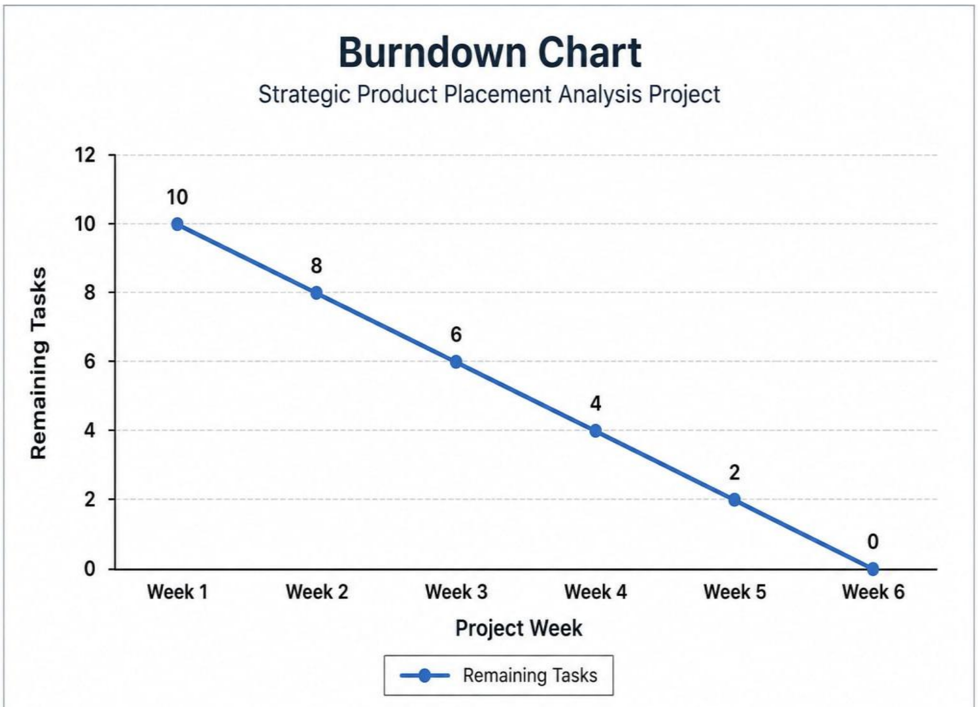
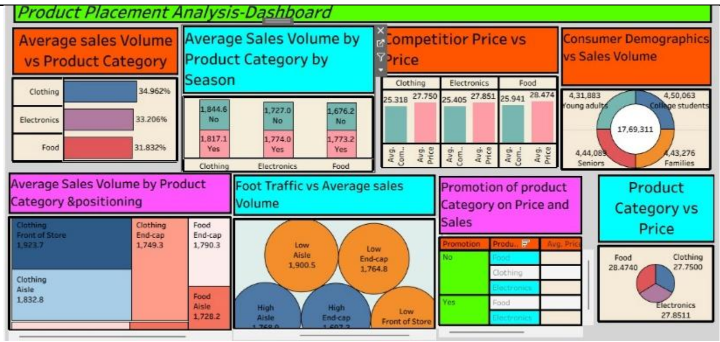
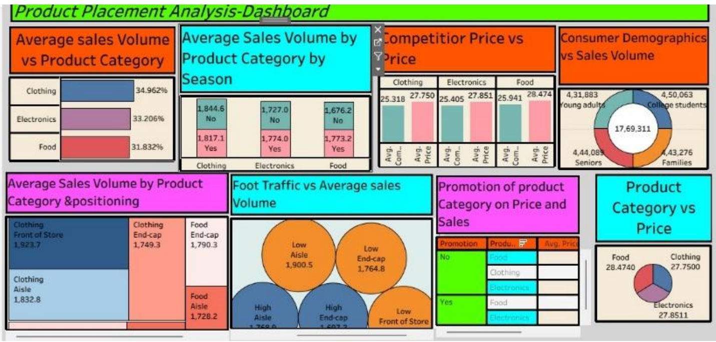
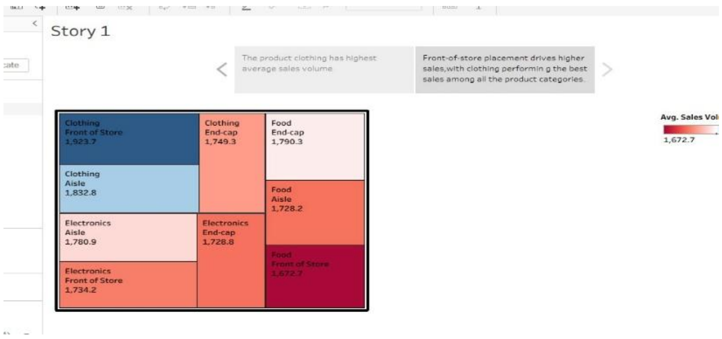


Figure: Burndown Chart for Strategic Product Placement Analysis Project

FUNCTIONAL AND PERFORMANCE TESTING:

Model Performance Testing:

Project team shall fill the following information in model performance testing template.

S.No.	Parameter	Screenshot / Values
1.	Data Rendered	
2.	Data Pre-processing	Removed null values, cleaned data, changed data types.
3.	Utilization of Filters	Measure Names(used to display only the required measures in the visualization).
4.	Calculation fields Used	Only one calculated field used i.e.zero
5.	Dashboard design	No of Visualizations / Graphs – 8 
6	Story Design	No of Visualizations / Graphs -2scenes 

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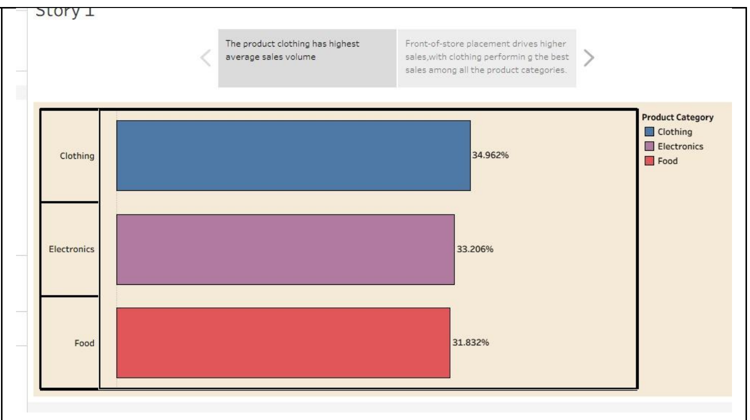
<tr><td></td><td></td><td>
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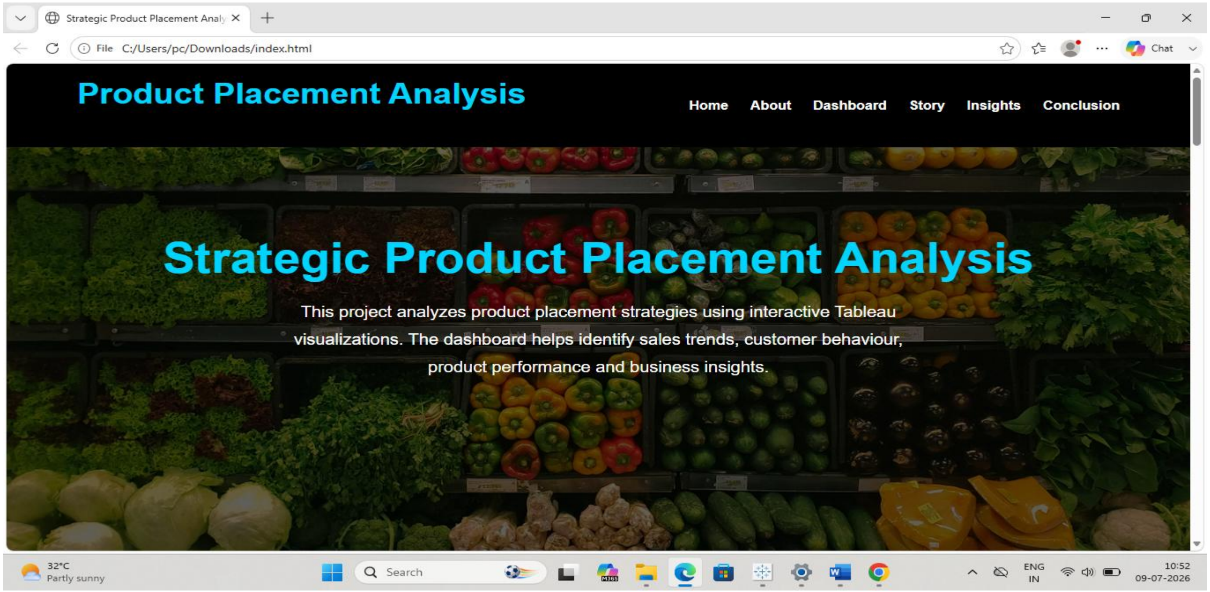
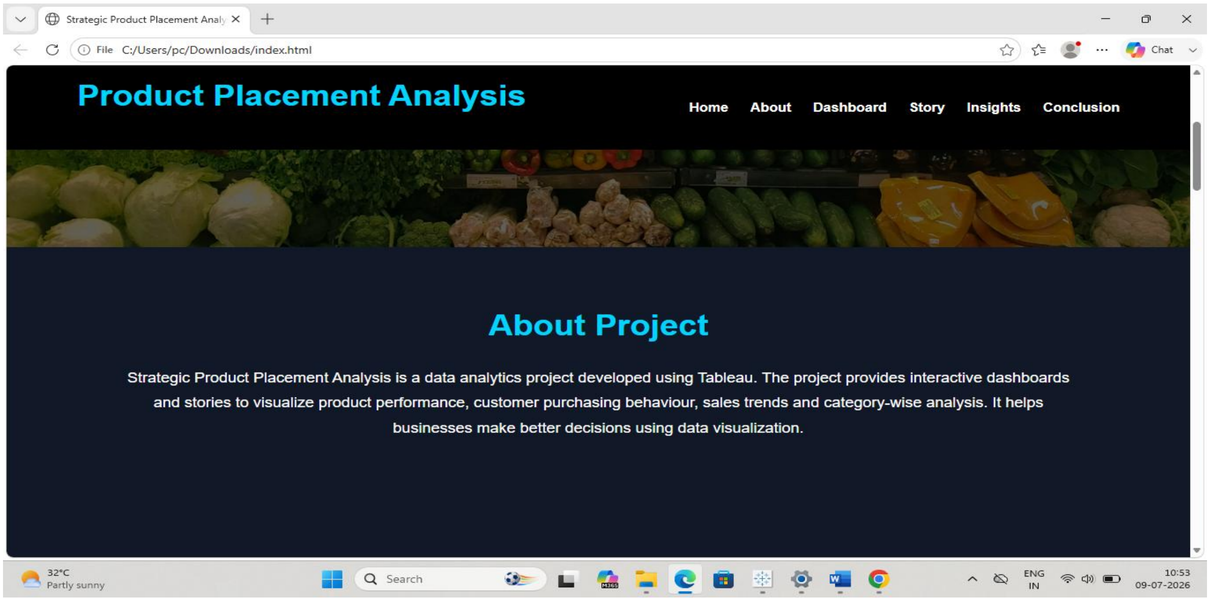
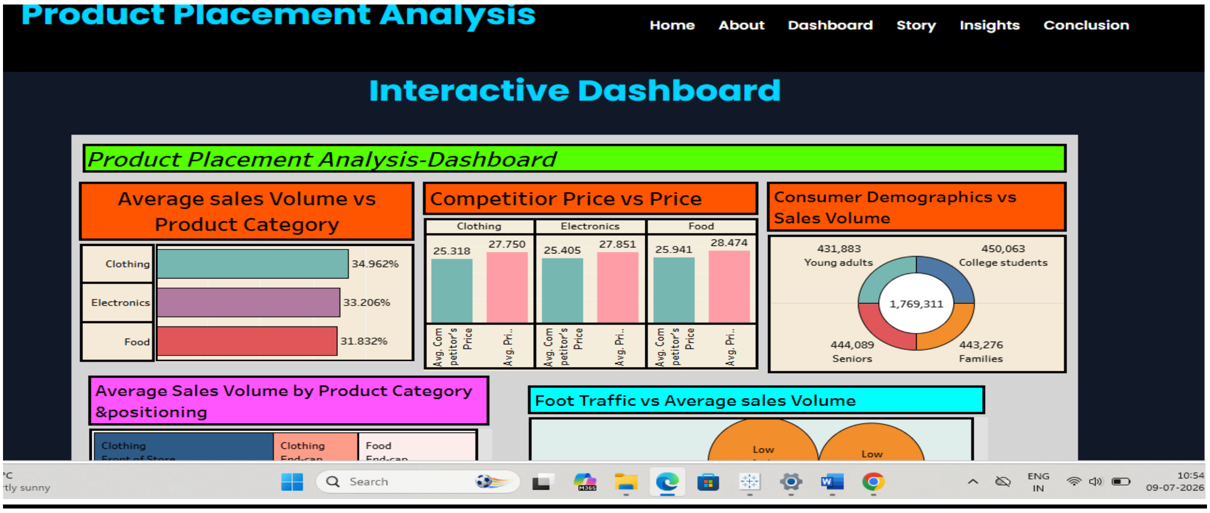
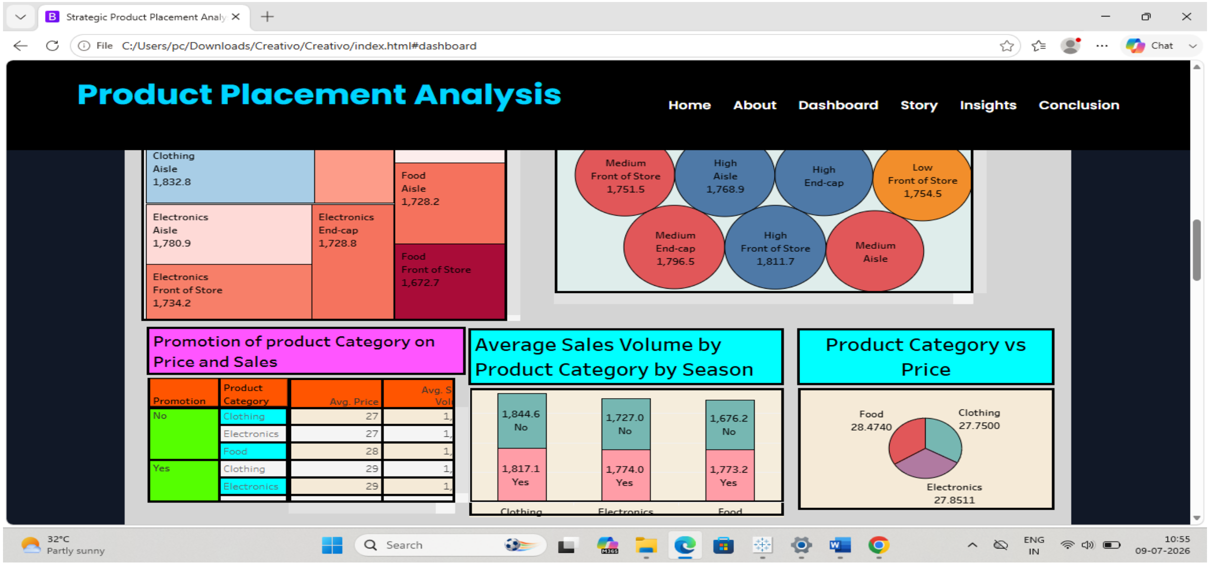
RESULTS:






Product Placement Analysis
Home About Dashboard Story Insights Conclusion
Interactive Dashboard
Product Placement Analysis-Dashboard
Average sales Volume vs Product Category
Clothing 34.962%
Electronics 33.206%
Food 31.832%
Competitior Price vs Price
Clothing Electronics Food
25.318 27.750 25.405 27.851 25.941 28.474
Consumer Demographics vs Sales Volume
431,883 Young adults
450,063 College students
1,769,311
444,089 Seniors
443,276 Families
Average Sales Volume by Product Category &positioning
Foot Traffic vs Average sales Volume
Search
10:54
09-07-2026


Strategic Product Placement Analy
File C:/Users/pc/Downloads/Creativo/Creativo/index.html#dashboard
Product Placement Analysis
Home About Dashboard Story Insights Conclusion
Clothing Aisle 1,832.8
Electronics Aisle 1,780.9
Electronics End-cap 1,728.8
Electronics Front of Store 1,734.2
Food Aisle 1,728.2
Food Front of Store 1,672.7
Medium Front of Store 1,751.5
High Aisle 1,768.9
High End-cap
Low Front of Store 1,754.5
Medium End-cap 1,796.5
High Front of Store 1,811.7
Medium Aisle
Promotion of product Category on Price and Sales
Promotion | Product Category | Avg. Price
No | Clothing | 27
Electronics | 27
Food | 28
Yes | Clothing | 29
Electronics | 29
Average Sales Volume by Product Category by Season
1,844.6 No
1,817.1 Yes
1,727.0 No
1,774.0 Yes
1,676.2 No
1,773.2 Yes
Clothing Electronics Food
Product Category vs Price
Food 28.4740
Clothing 27.7500
Electronics 27.8511
32°C Partly sunny
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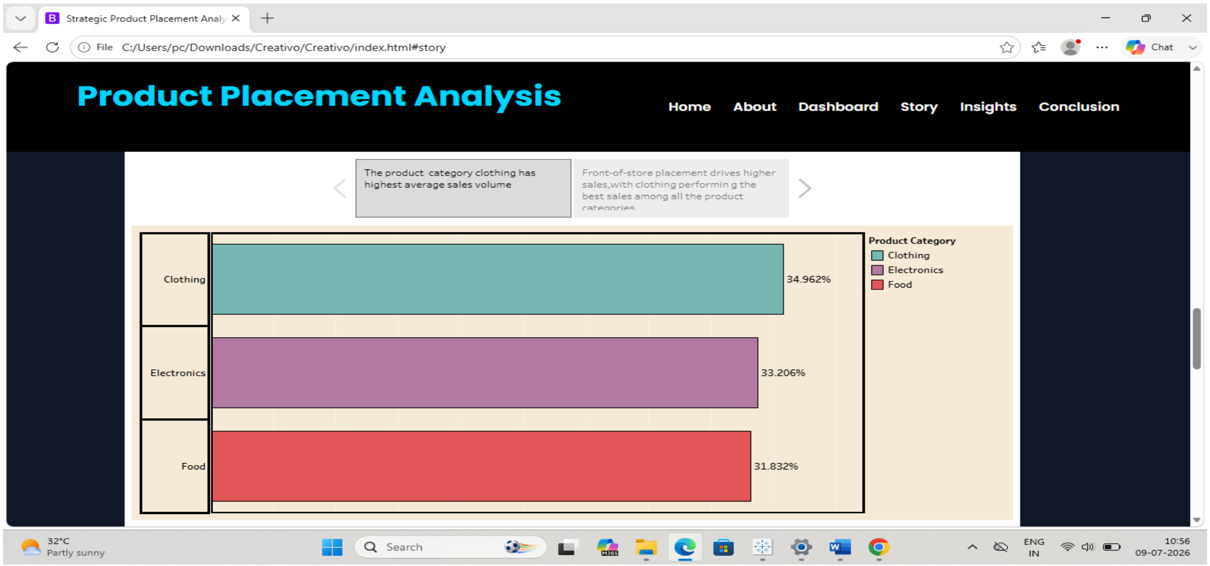
Strategic Product Placement Analy
File C:/Users/pc/Downloads/Creativo/Creativo/index.html#story
Product Placement Analysis
Home About Dashboard Story Insights Conclusion
The product category clothing has highest average sales volume
Front-of-store placement drives higher sales,with clothing performin g the best sales among all the product categories
Clothing 34.962%
Electronics 33.206%
Food 31.832%
Product Category
Clothing
Electronics
Food
32°C Partly sunny
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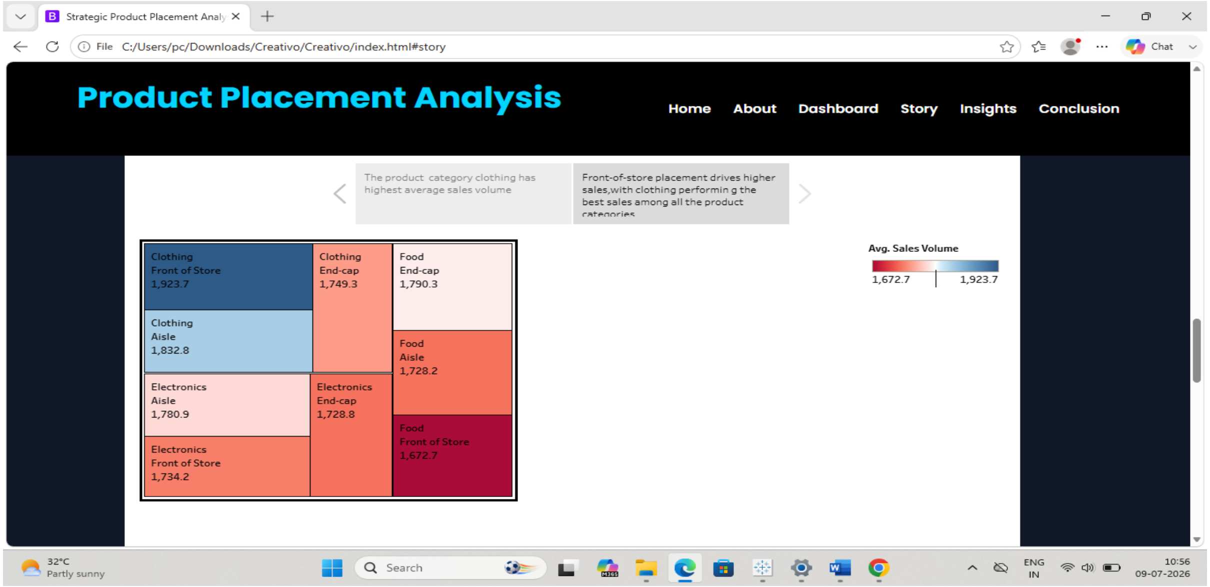
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Product Placement Analysis
Home About Dashboard Story Insights Conclusion
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Clothing Front of Store 1,923.7
Clothing End-cap 1,749.3
Food End-cap 1,790.3
Clothing Aisle 1,832.8
Food Aisle 1,728.2
Electronics Aisle 1,780.9
Electronics End-cap 1,728.8
Food Front of Store 1,672.7
Electronics Front of Store 1,734.2
Avg. Sales Volume
1,672.7
1,923.7
32°C Partly sunny
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ENG IN
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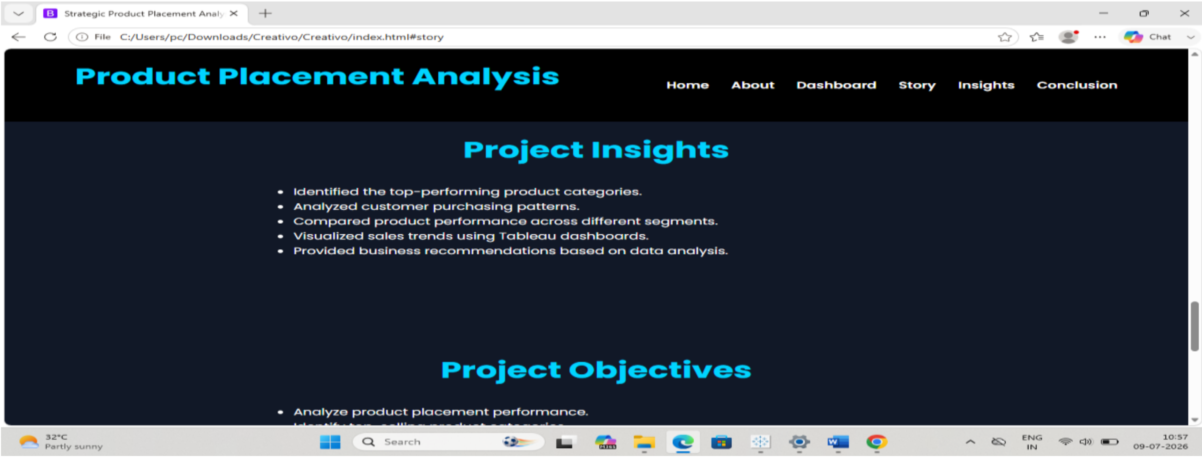
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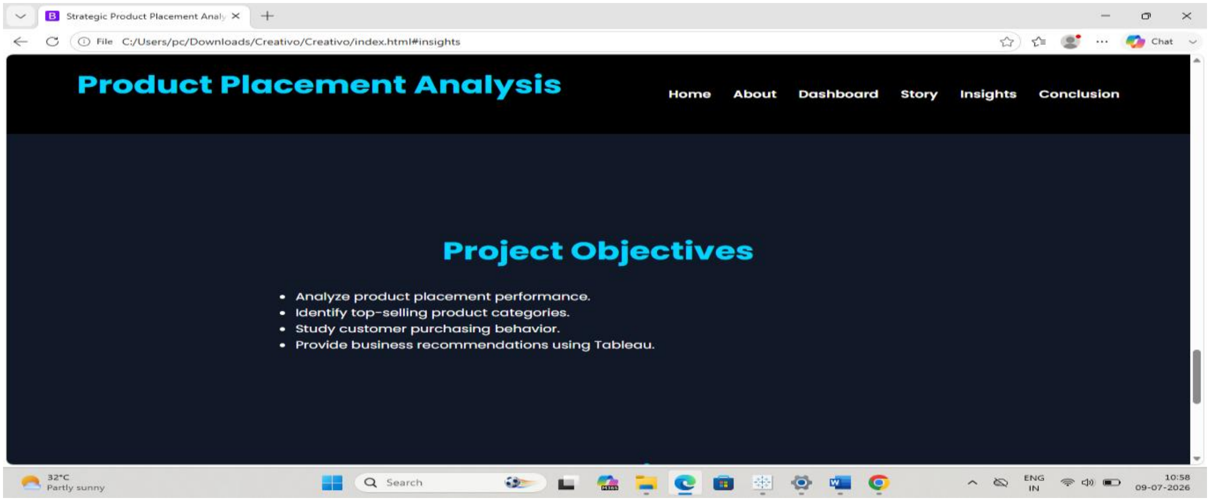
Product Placement Analysis
Home About Dashboard Story Insights Conclusion
Project Insights
Identified the top-performing product categories.
Analyzed customer purchasing patterns.
Compared product performance across different segments.
Visualized sales trends using Tableau dashboards.
Provided business recommendations based on data analysis.
Project Objectives
Analyze product placement performance.
32°C Partly sunny
Search
ENG IN
10:57 09-07-2026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Product Placement Analysis
Home About Dashboard Story Insights Conclusion
Project Objectives
Analyze product placement performance.
Identify top-selling product categories.
Study customer purchasing behavior.
Provide business recommendations using Tableau.
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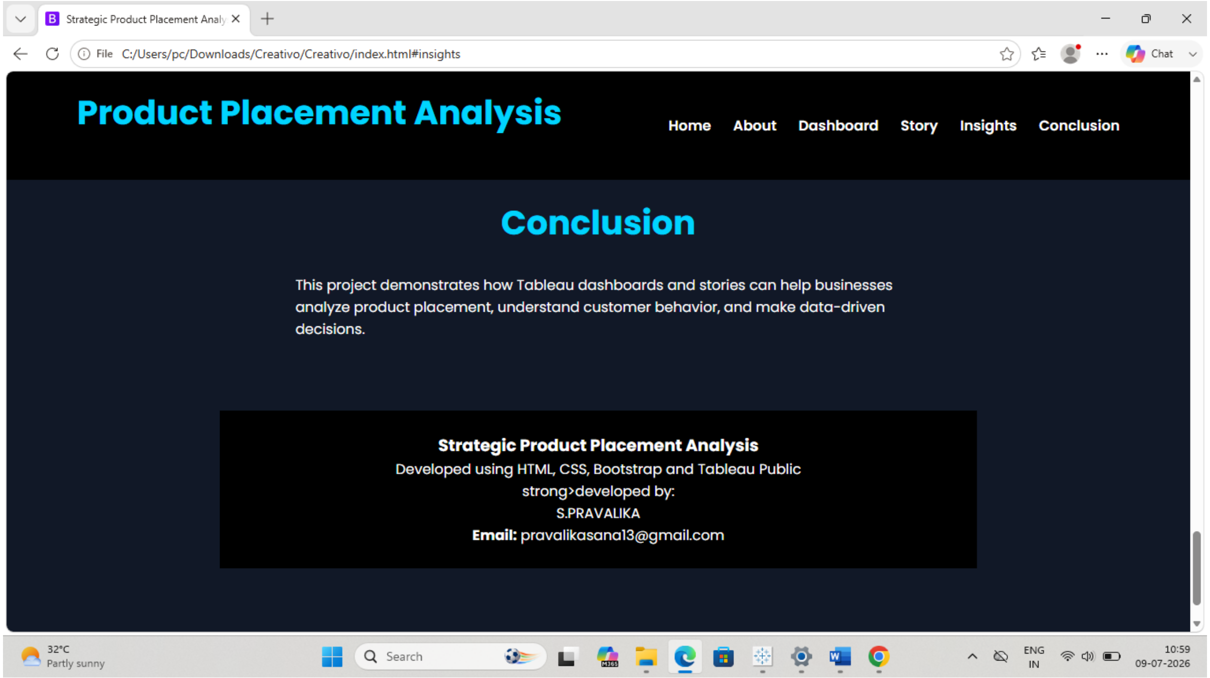

<u>ADVANTAGES & DIS-ADVANTAGES:</u>

Advantages:

- Helps identify the best product placement strategies.
- Improves customer shopping experience.
- Increases sales and profitability through data-driven insights.
- Provides interactive dashboards for quick decision-making.
- Saves time by automating data analysis and reporting.
- Helps compare sales performance across products, regions, and categories.
- Supports better inventory and merchandising decisions.

Disadvantages:

- Accuracy depends on the quality of the input data.
- Requires regular data updates to provide current insights.
- Tableau Public dashboards may expose data if not handled carefully.
- Does not predict future sales without advanced predictive models.
- Requires basic knowledge of Tableau to create and maintain dashboards.
- External factors (seasonality, promotions, market trends) may affect sales but are not always captured in the analysis.

Applications:

- Retail Stores
- Supermarkets
- Business Intelligence
- Inventory Management etc…

CONCLUSION:

The Strategic Product Placement Analysis project demonstrates how data analytics can help retailers optimize product placement and improve business performance. By analyzing sales data using Tableau, the project provides interactive dashboards that highlight sales trends, customer purchasing behaviour, and product performance. These insights enable retailers to make informed decisions, enhance customer satisfaction, improve inventory management, and increase overall sales. The project shows that data-driven product placement strategies can significantly support efficient retail operations and better business growth.

FUTURE SCOPE:

The Strategic Product Placement Analysis project can be enhanced in several ways in the future:

- Integrate real-time sales data for live analysis.
- Apply Machine Learning algorithms to predict future sales and customer buying behaviour.
- Develop product placement recommendations using Artificial Intelligence (AI).
- Connect with cloud databases for automatic data updates.
- Create a mobile-friendly dashboard for easy access by managers.
- Expand the system to support multiple retail stores and larger datasets.
- Include customer feedback and market trend analysis for more accurate recommendations.

APPENDIX:

Source code:

C:\Users\pc\Downloads\Creativo\Creativo\index.html

Strategic Product Placement Analysis

https://github.com/pravalika-Sana/Strategic-Product-Placement-Analysis/blob/main/project%20code%20file/index.html

Demo link:

https://github.com/pravalika-Sana/Strategic-Product-Placement-Analysis/blob/main/videodemo/Demo.mp4

Demo.mp4

GITHUB LINK:

https://github.com/pravalika-Sana/Strategic-Product-Placement-Analysis